Since AI 2026 · hackathon project

Fire & life safety · SOLAS II-2 · FSS Code

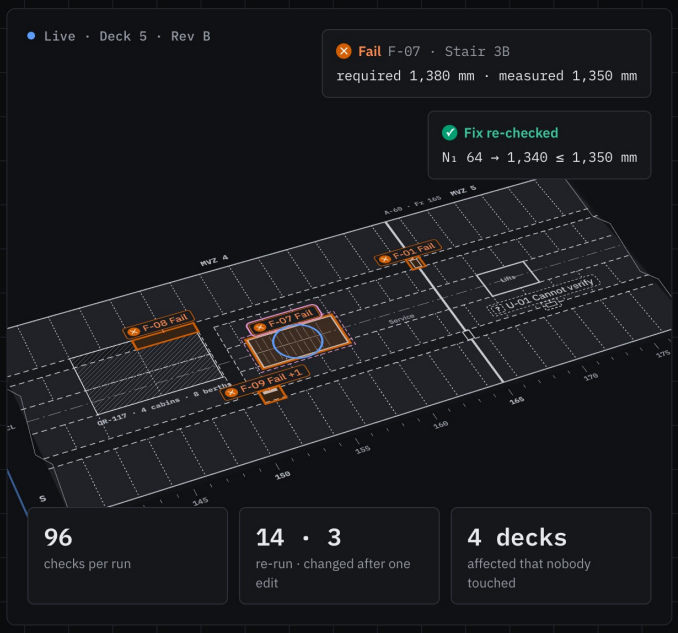
Ship safety, checked while you design.

SonarIQ finds the fire-safety rule a design change breaks, shows it on the model with the rule's own numbers, and offers fixes it has already re-checked.

Open live demo → ▶ Watch the 2-min demo

No account needed IFC4 in · BCF out

Rule engine decides · AI explains



Checks against SOLAS II-2 FSS Code Ch.13 PVAG 2013 (draft) ADA 2010 DNV class · to verify

Works with FreeCAD · IFC4 · BCF

How it works

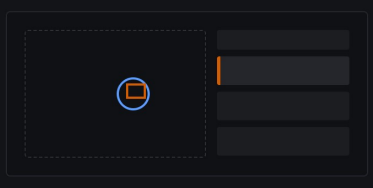
One change. Every rule it touches. Seconds later.

Four cabins added aft of frame 150 make a stair four decks away too narrow. SonarIQ catches it before anyone opens a report.

01 · In FreeCAD

Analyse where you design

A docked panel checks the open IFC model. Findings light up on the 3D model first, then in a list.



02 · Evidence

Proof, not a score

Every verdict shows the clause, required against measured, and the formula filled with the model's own inputs.

$$W = (N_1 + N_2 + 0.5\,N_3 + 0.25\,N_4) \times 10$$
$$= (68 + 44 + 20 + 6) \times 10$$
$$= 1{,}380 \text{ mm required}$$
$$\times\ 1{,}350 \text{ mm measured} \cdot -30 \text{ mm}$$

03 · Review portal

Decide, sign, hand back

Discipline leads approve, request changes or decline. Every action is signed. Issues go back to any CAD tool as BCF.

- Rule engine found F-07 — 14:32
- Designer applied fix 2 — 14:36
- Fire & life safety lead approved — 15:06

Approve · signed

Trust by design

The engine decides. The engineer decides. AI explains.

A deterministic rule engine sets every Pass and Fail. AI only explains in plain words, turns typed requests into edits you confirm, and proposes fixes the engine re-checks before you see them. When the model is missing data, SonarIQ says “Cannot verify”. It never guesses.

Rule engine

Sets every verdict

Same input, same answer. Tested rule by rule.

✓ AI explanation · numbers checked

Never sets a status

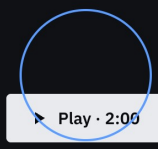
Uses only numbers the engine produced.

Fail · Warning · ? Cannot verify · Pass · Accepted with reason

Every status has an icon and a word. Colour-blind safe, WCAG 2.1 AA.

Demo video

See the golden path in two minutes



Play · 2:00

Analyse → typed change → ping → fix → review · poster uses the logo animation

Built at Since AI 2026

A 36-hour prototype for the Meyer Turku challenge

The challenge: automated class and regulatory compliance checking for cruise ship design. SonarIQ is our answer, a working prototype rather than a product. Rules marked “To verify” have not been checked by a class society.

Hackathon project · Not class approved · Open source · [licence]

36 h
from brief to demo

96
checks in 3 rule packs

2
ship models: demo ship and RCRV

Team [team member names and roles] Turku · [date]

Resources

Everything we submitted

Live demo

Open the demo workspace

Portal with the demo ship and RCRV. No account needed.

34-88-130-224.sslip.io/next →

Code

GitHub repository

Rule engine, FreeCAD add-on, portal and tests.

github.com/[org]/sonariq →

Video

Demo video · 2 min

The golden path from Analyse to a signed approval.

[demo video URL] →

Deck

Pitch deck

Problem, approach, results and what comes next.

[pitch deck URL] →

Since AI

Hackathon submission

Our entry for the Meyer Turku challenge.

[Since AI submission URL] →

Docs

Spec and design hand-off

Product spec, rulebook and the UI design canvas.

[docs URL] →

Review in the browser. Fix in your CAD tool.

Open live demo →

Get the FreeCAD add-on

View on GitHub